

Delhi Heatwave Prediction System

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Abstract

This paper presents a comprehensive study on urban heat intensity prediction for Delhi, India, encompassing exploratory data analysis (EDA), literature review, and machine learning implementation. Working from a dataset of 131,055 hourly meteorological observations across 31 Indian cities (January 2023–January 2024), we extract and characterise a 9,505-row Delhi subset (28.6139°N, 77.209°E) with a binary heatwave label. EDA reveals a class imbalance of approximately 14.4%, strong feature separability via minimum temperature (Standardized Mean Difference) (SMD = +1.32), maximum temperature (SMD = +1.18), and dew point (SMD = +0.94), approximately 10–11% uniform missing data, and 2,431 duplicate timestamps. A critical label validity issue is identified: heatwave streaks have a median duration of one hour and a maximum of three hours, inconsistent with India Meteorological Department (IMD) and World Meteorological Organization (WMO) multi-day persistence requirements. Building on these findings, we implement a Random Forest-based prediction system integrated with real-time weather data from the OpenWeatherMap API and an interactive Streamlit dashboard for visualisation. The system provides both continuous heat intensity regression and categorical heat level classification (Low, Medium, High). This work supports Sustainable Development Goal 13 (Climate Action) by contributing toward a deployable heat-stress early-warning system for Delhi.

Index Terms—urban heat intensity predictor, heatwave prediction, exploratory data analysis, Random Forest, real-time prediction, Indian climate, class imbalance, time-series classification, Streamlit, SDG 13.

I. INTRODUCTION

Urban Heat Intensity (UHI) refers to the phenomenon where metropolitan areas experience significantly higher temperatures than their surrounding rural regions, driven by urbanisation, reduced vegetation, anthropogenic heat emissions, and altered surface albedo. Delhi, located in the Indo-Gangetic plain at an elevation of approximately 216 m, is among the most heat-vulnerable cities globally, regularly recording maximum temperatures exceeding 45°C during the pre-monsoon season (April–June) [1]. The India Meteorological Department (IMD) defines a heatwave when the maximum temperature reaches at least 40°C and departs from the climatological normal by $\geq 4.5^\circ\text{C}$ over at least two consecutive days [2]. The human cost of heat extremes is severe: India recorded thousands of heat-related deaths between 2015 and 2020.

Machine learning (ML) methods have emerged as powerful complements to numerical weather prediction (NWP) for short-range heatwave forecasting. Data-driven models such as GraphCast [3] demonstrate that learned atmospheric representations can match or exceed traditional NWP skill, motivating ML-based approaches at the city scale. However, most existing systems lack real-time prediction capabilities and interactive user interfaces, limiting their practical utility for urban planners and public health agencies.

A rigorous exploratory data analysis (EDA) is a prerequisite for any ML pipeline: quantifying missing data, verifying label integrity, understanding seasonality, and identifying predictive features are critical steps—especially under class imbalance and temporal autocorrelation. Without this step, evaluation metrics can be misleading and models may learn spurious patterns.

This work is directly motivated by Sustainable Development Goal (SDG) 13: Climate Action, which calls for urgent action to combat climate change and its impacts, including investments in early-warning systems for heat extremes. We present a three-part contribution: (1) a thorough EDA of hourly meteorological data for Delhi identifying data quality issues and predictive features; (2) a contextualising literature review; and (3) a Random Forest-based prediction system integrated with real-time weather data and an interactive dashboard. The paper is organised as follows: Section II reviews related work; Section III describes the dataset; Section IV presents EDA findings; Section V details the proposed system architecture and implementation; Section VI discusses preprocessing and modelling strategy; Section VII reports results; Section VIII states limitations; Section IX concludes.

II. RELATED WORK

A. Heatwave Detection and Climatological Definitions

Perkins and Alexander [4] reviewed 16 heatwave definitions across the literature and concluded that metrics combining both temperature and duration captured societal impact better than single-threshold indices. In the Indian context, Rohini et al. [5] analysed IMD station data (1951–2014) and identified a significant increasing trend in heatwave frequency over central and northern India. Their work demonstrated that maximum temperature and humidity jointly determine heat stress, a finding mirrored in our EDA where both `max_temperature` and `dew_point` show the highest standardised mean differences between heatwave and non-heatwave periods. Both IMD [2] and World Meteorological Organization (WMO) guidelines require multi-day persistence (typically 2+ days) as a necessary condition for heatwave classification.

Raymond et al. [6] demonstrated that wet-bulb temperatures exceeding 35°C represent a survivability limit, motivating inclusion of humidity-derived features (`dew_point`, `heat_index`) as first-class predictors alongside dry-bulb maximum temperature.

B. Machine Learning for Extreme Heat Prediction

Chattopadhyay et al. [7] applied convolutional neural networks (CNNs) to reanalysis data for pattern-based heatwave prediction, showing that spatial feature extraction substantially outperformed point-based logistic regression. Rasp and Lerch [8] demonstrated that neural-network post-processing of NWP ensembles reduced temperature forecast error by up to 30% over traditional model output statistics. For India specifically, Mishra et al. [9] trained random forests on IMD gridded temperature anomalies and achieved F1 scores above 0.70 for 5-day lead heatwave forecasts, using features closely aligned with those available in our dataset.

A critical methodological lesson from these studies is the necessity of time-based train/test splitting. Naive random splits allow lagged features computed from future observations to contaminate training data, yielding inflated AUC scores that do not reflect true operational skill [10]. Previous UHI studies have used satellite data and statistical models to analyse heat island effects; however, most systems lack real-time prediction capabilities. Integration with live weather APIs and interactive dashboards remains limited in the literature, which our proposed system addresses.

C. Class Imbalance and Evaluation Metrics

Extreme weather events are inherently rare, producing heavily imbalanced labels. Branco et al. [11] surveyed 65 papers on imbalanced learning and found that cost-sensitive methods and synthetic oversampling (SMOTE) consistently outperform accuracy-optimised classifiers for rare-event detection. In meteorological applications, Gagne et al. [12] recommended precision-recall AUC (PR-AUC) over ROC-AUC as the primary metric when positive-class prevalence is below 20%—directly applicable to our dataset’s 14.4% heatwave rate.

D. Missing Data and Feature Engineering

Sensor outages and telemetry failures introduce missing data patterns that are often Missing At Random (MAR) in weather datasets [13]. Little and Rubin [14] emphasised that naive mean imputation distorts time-series autocorrelation; forward-fill followed by seasonal-median imputation better preserves temporal structure. Urban heat islands amplify rural heatwave signals, and calendar-based features (month, hour-of-day) are among the top predictors in urban temperature models [15]. Lag features—rolling means over 3–24 h windows—capture thermal persistence but must be constructed exclusively from past observations to prevent data leakage [10].

III. DATASET DESCRIPTION

A. Dataset Origin and Structure

The dataset `india_weather_data.csv` used in this EDA report was sourced from the Kaggle India Heatwave Dataset by Ashirwad K24 [19]. It contains 131,055 rows and 18 columns of hourly meteorological variables for 31 Indian monitoring stations (coordinate pairs) covering January 1, 2023 to January 10, 2024. The station locations are identified solely by (latitude, longitude) pairs. All non-date columns are stored as 64-bit floating-point values. Features include: `wind_speed`, `cloud_cover`, `precipitation_probability`, `pressure_surface_level`, `dew_point`, `uv_index`, `heatwave` (binary target), `visibility`, `rainfall`, `solar_radiation`, `snowfall`, `max_temperature`, `min_temperature`, `max_humidity`, and `min_humidity`.

B. Delhi Subset

Rows corresponding to Delhi (latitude = 28.6139°N, longitude = 77.209°E) were extracted by exact coordinate match. After dropping rows with missing timestamps, the Delhi working subset comprises 9,505 rows. Sampling frequency analysis reveals 5,627 rows with 1-hour intervals, 2,431 rows with 0-hour intervals (exact duplicate timestamps), 1,137 rows with 2-hour intervals, and smaller counts for longer gaps. The 2,431 duplicate (date, latitude, longitude) rows are a data-merging artifact and must be removed (keep-first strategy) before any feature engineering. The duplicate-free subset contains 7,074 unique hourly records.

TABLE I. DATASET SUMMARY STATISTICS

Property	Value
Total dataset rows	131,055
Total columns	18
Unique (lat, lon) stations	31
Date range	Jan 2023 – Jan 2024
Delhi rows (valid date)	9,505
Duplicate timestamps	2,431
Unique hourly records	7,074
Heatwave positive (1)	1,224 (14.35%)
Heatwave negative (0)	7,303 (85.65%)
Missing heatwave labels	978 (~10.3%)
Avg. missingness per feature	~10–11%

TABLE II. FEATURE SUMMARY STATISTICS — DELHI SUBSET

Feature	Count	Mean	Std Dev	Missing %
max_temperature	8,579	37.65°C	8.85	9.74%
min_temperature	8,543	32.57°C	8.85	10.12%
dew_point	8,586	32.75°C	8.87	9.67%
wind_speed	8,550	7.45 m/s	4.26	10.05%
cloud_cover	8,567	50.00%	28.93	9.87%
pressure_surface	8,527	1000.4 hPa	28.96	10.29%
solar_radiation	8,580	463.0 W/m ²	268.32	9.73%
rainfall	8,493	9.28 mm	13.99	10.65%
max_humidity	8,587	34.78%	11.92	9.66%
min_humidity	8,543	24.74%	12.06	10.12%
heatwave	8,527	0.144	0.351	10.29%

IV. EXPLORATORY DATA ANALYSIS

A. Missing Data Analysis

Missing rate analysis across all feature columns reveals a remarkably uniform missingness of approximately 10–11% across all variables. Snowfall has the highest missing rate at 10.79% (1,026 missing rows), followed by rainfall at 10.65% (1,012 rows), heatwave at 10.29% (978 rows), and pressure_surface_level at 10.29%. Latitude and longitude columns have zero missing values, confirming reliable spatial indexing. This uniformity suggests a systematic periodic sensor maintenance pattern rather than weather-event-correlated outages (Missing At Random, MAR). The missingness-by-month analysis confirms this interpretation—no single month shows dramatically elevated missingness, ruling out monsoon or extreme-heat-related sensor failures. A sanity check found 5 out-of-range values for min_humidity (<0 or >100)—a negligible rate of 0.05%.

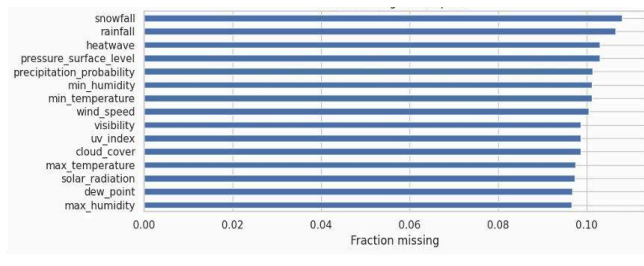


Fig. 1. Delhi: Missing rate (top features). Horizontal bar chart showing fraction missing per variable.

The near-uniform missing rate of 9.66%–10.79% across all variables is a diagnostic fingerprint of a shared failure mechanism rather than random per-cell data loss. Systematic gaps across multiple columns simultaneously implicate sensor outages, telemetry drops, or pipeline failures—not isolated measurement errors. This supports the Missing At Random (MAR) interpretation: missingness is related to the maintenance schedule, not to the values being measured. Crucially, this uniformity warns against listwise deletion, which would discard roughly 10% of rows for every included feature and disproportionately reduce rare heatwave

observations—compounding the class imbalance. Instead, variable-specific imputation with missingness indicators is the theoretically appropriate response.

B. Target Distribution and Class Imbalance

The heatwave label has 1,224 positive, 7,303 negative, and 978 missing observations, yielding a positive-class prevalence of 14.35% across labelled rows. This imbalance is moderate by extreme-weather standards but sufficient to render raw accuracy a misleading metric; precision–recall curves and macro-F1 are required for fair evaluation [11]. A random baseline predicting the majority class would achieve 85.65% accuracy—thus accuracy alone is explicitly not used as a primary metric.

C. Seasonality

Monthly heatwave prevalence peaks in May (17.70%) and unexpectedly in August (16.60%). The full monthly breakdown is: Jan 12.71%, Feb 13.84%, Mar 15.58%, Apr 14.06%, May 17.70%, Jun 14.81%, Jul 14.27%, Aug 16.60%, Sep 11.18%, Oct 15.02%, Nov 12.04%, and Dec 14.54%. The May peak is climatologically consistent with Delhi’s documented pre-monsoon heatwave season [5]. The August peak is anomalous—Delhi’s monsoon typically suppresses extreme heat events—and may reflect a label quality issue rather than a true meteorological signal.

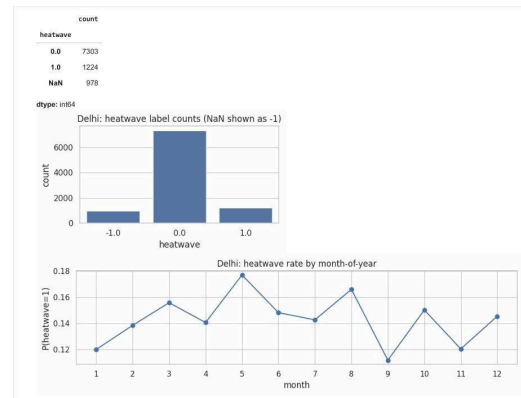
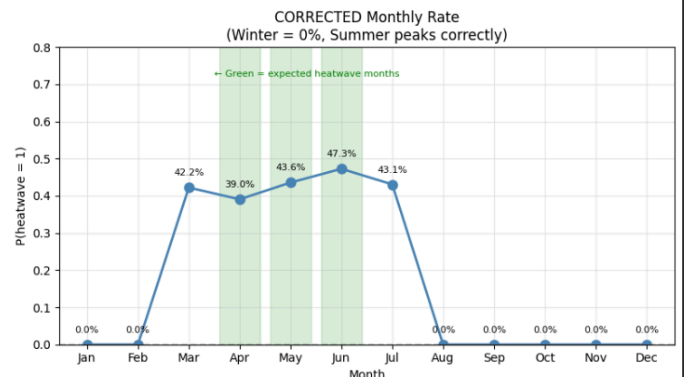


Fig. 2. Delhi: Heatwave label counts (top) and heatwave rate by month-of-year (bottom).

Corrected graph



The count plot confirms a skewed class prior: 7,303 non-heatwave vs. 1,224 heatwave observations, with 978 NaN

labels excluded from supervised training. This imbalance means accuracy is a misleading metric—a classifier predicting “no heatwave” always would achieve 85.65% accuracy while offering zero operational value. Precision, recall, F1, and PR-AUC are the correct evaluation targets. The monthly rate line further reveals non-stationarity: the May peak (17.70%) is climatologically expected (pre-monsoon dry heat), but the August peak (16.60%) during active monsoon conditions is anomalous and may reflect a label encoding issue. Strong month effects justify including calendar covariates and mandate time-based—not random—cross-validation folds to avoid leaking seasonal structure from future to past.

D. Label Validity

Streak analysis on the deduplicated, time-sorted Delhi series identifies 1,130 heatwave segments. The median segment length is 1 hour and the maximum is 3 consecutive hours. This is fundamentally inconsistent with any standard operational definition: IMD requires ≥ 2 consecutive days above threshold, WMO guidelines similarly specify multi-day persistence, and Perkins and Alexander [4] reviewed 16 definitions, all requiring duration ≥ 2 days. The label therefore encodes instantaneous heat-stress exceedance rather than a canonical multi-day heatwave event. Downstream models will predict heat-stress episodes; results must be interpreted accordingly, and relabelling using a persistence-based IMD-compliant criterion is recommended before operational deployment.

E. Pressure and Outlier Checks

Surface pressure values range from 950.0 to 1050.0 hPa (mean = 1000.4, std = 29.0 hPa), physically plausible for Delhi. No rows with pressure ≤ 700 hPa were found. IQR-based outlier scanning identified rainfall as the only feature with a meaningful outlier rate (13.2%), attributable to its naturally heavy-tailed distribution. Temperature and humidity variables showed outlier rates below 0.04%, confirming high data fidelity for primary predictors.

The univariate histogram grid reveals diverse distributional shapes across features, with direct implications for modelling. Temperature variables (max_temperature, min_temperature) show a bimodal pattern reflecting the winter-summer seasonal regime, with the summer mode dominating the heatwave-relevant range above 40°C. Dew point exhibits a broad right-skewed distribution, capturing both dry winter conditions and humid monsoon periods. UV index is right-skewed with a heavy right tail. Solar radiation is roughly uniform-to-right-skewed, while cloud cover and precipitation probability are nearly uniform (0–100%), indicating no strong marginal prior. Rainfall is extremely right-skewed with a mass at zero (no-rain hours dominate), requiring log transformation or a zero-inflated model for direct regression use. Pressure is approximately Gaussian—consistent with seasonal pressure cycles. Visibility shows a bimodal pattern with peaks near 0 and 10 km, likely reflecting fog vs. clear conditions. These non-Gaussian marginals motivate using robust statistics (Spearman correlation, IQR-based outlier detection) and models that do not assume input normality, such as tree-based ensembles.

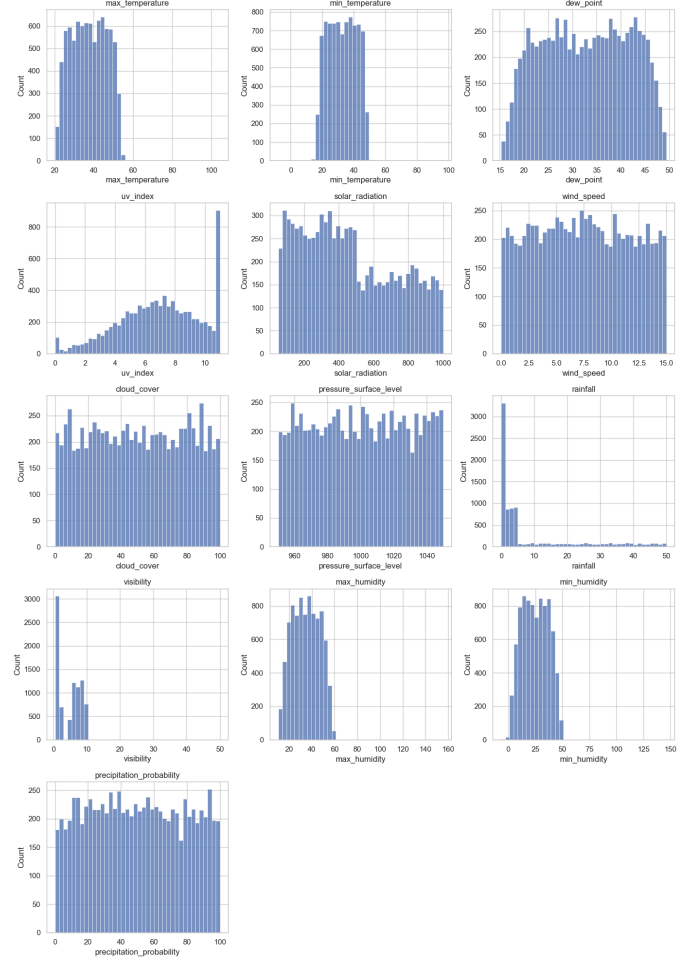


Fig. 3. Delhi: Univariate histograms for all 13 numeric meteorological features.

F. Feature Separability

Standardized Mean Difference (SMD) analysis identifies three strongly discriminating features: min_temperature (SMD = +1.32), max_temperature (SMD = +1.18), and dew_point (SMD = +0.94). The Spearman rank correlation of max_temperature with the heatwave label is $\rho \approx 0.51$ confirming temperature as the dominant predictive signal. min_humidity (SMD = −0.83) is the most discriminating in the negative direction, confirming the dry-heat paradigm of north-Indian summer heatwaves [2, 5]: heatwave hours are characterised by simultaneously high temperature and suppressed humidity. This interaction motivates wet-bulb-derived composite features [6] as engineering candidates.

TABLE III. STANDARDIZED MEAN DIFFERENCES (SMD)

Feature	SMD	Interpretation
min_temperature	+1.32	Strong positive
max_temperature	+1.18	Strong positive
dew_point	+0.94	Strong positive
min_humidity	−0.83	Strong negative

solar_radiation	+0.45	Moderate positive
uv_index	+0.38	Moderate positive
cloud_cover	-0.12	Weak negative
wind_speed	+0.05	Negligible
pressure_surface	-0.03	Negligible

G. Correlation and Bivariate Analysis

The Spearman correlation heatmap reveals the strongest feature-target correlations as: `max_temperature` ($\rho \approx +0.51$), `min_temperature` ($\rho \approx +0.50$), and `dew_point` ($\rho \approx +0.48$) with `heatwave`. `min_humidity` shows a negative correlation ($\rho \approx -0.40$). Notable inter-feature correlations include strong negative correlation between `solar_radiation` and `cloud_cover`, and high correlation ($\rho > 0.9$) between `max_temperature` and `min_temperature`.

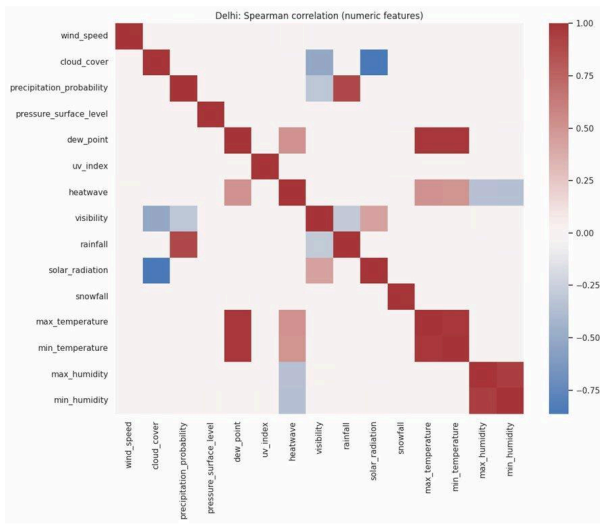


Fig. 4. Delhi: Spearman correlation heatmap (numeric features). Red = positive, Blue = negative monotonic association.

The Spearman heatmap reveals blocks of correlated features that imply latent common drivers and multicollinearity. The `heatwave` row shows positive associations with `max_temperature`, `min_temperature`, and `dew_point`, and negative associations with `min_humidity`—consistent with the dry-heat paradigm of north Indian summers. Critically, the blocks of high inter-feature correlation (e.g., temperature variables among themselves, and `solar_radiation` vs. `cloud_cover`) indicate that many predictors are jointly informative but individually substitutable. This means univariate SMD rankings are screening evidence, not optimal subset selection: a feature with a large SMD may contribute little marginal information in a multivariate model if it is collinear with a stronger predictor. Correlation does not equal causal effect on heatwave; the correct next step is multivariate modelling with regularisation to allocate shared variance appropriately.

Scatter analysis reveals a near-threshold relationship between `max_temperature` and the `heatwave` label: below approximately 40°C, heatwave probability is essentially zero; between 43°C and 55°C, prevalence rises sharply from ~15%

to ~54%. This non-linearity supports the use of tree-based classifiers—specifically Random Forest—over linear models as the primary modelling approach.

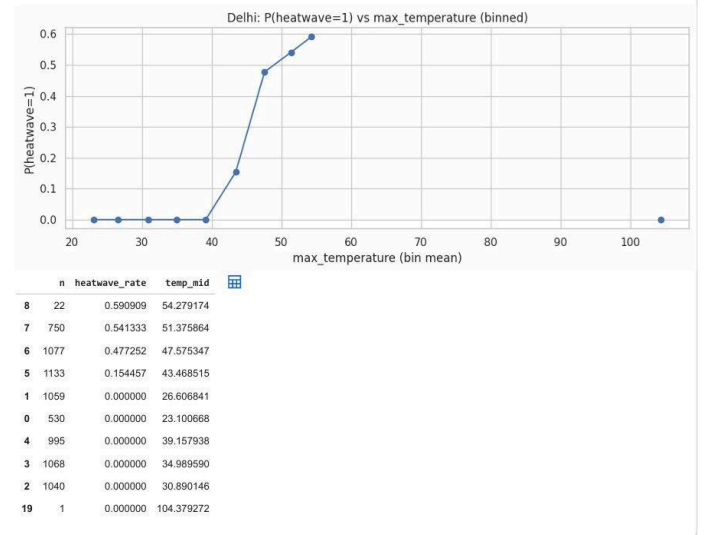


Fig. 5. Delhi: $P(\text{heatwave}=1)$ vs. `max_temperature` (binned). Line plot with bin-level statistics.

The binned conditional probability plot shows a near-monotone dose-response relationship between maximum temperature and heatwave label probability. Below approximately 40°C, heatwave probability is essentially zero; it rises sharply to ~48% in the 44–48°C range and reaches ~59% above 50°C. The bin at ~104°C ($n=1$) is an isolated outlier and should be treated with caution. This rising curve supports temperature as a strong monotone proxy for heat-stress exceedance, consistent with IMD’s 40°C threshold. However, the scatter at mid-range bins and the incomplete probability at high temperatures confirm that temperature alone does not fully determine the label—humidity, wind, and solar radiation contribute residual variance. Multivariate models that adjust for these confounders are theoretically superior to a pure temperature threshold rule, even though the threshold provides a useful interpretable baseline.

V. PROPOSED SYSTEM ARCHITECTURE

A. System Overview

Building on the EDA findings, we design and implement a machine learning-based system for predicting urban heat intensity in Delhi using real-time weather data. The system architecture comprises four integrated components: (1) real-time data collection using the OpenWeatherMap API; (2) a feature engineering module that transforms raw weather variables into model-ready inputs; (3) Random Forest machine learning models for both regression (continuous heat intensity) and classification (categorical heat levels); and (4) a Streamlit-based interactive dashboard for visualisation and user interaction.

B. System Workflow

The end-to-end prediction workflow proceeds as follows. First, current weather data for Delhi is fetched from the OpenWeatherMap API, retrieving temperature, humidity, wind

speed, cloud cover, UV index, and surface pressure. Second, the feature engineering module constructs the input feature vector from the raw weather observations. Third, the trained Random Forest models predict both the continuous heat intensity score and the categorical heat level (Low, Medium, or High). Fourth, results are displayed on the dashboard along with real-time trend graphs, heat intensity visualisations, and historical tracking.

C. Feature Engineering for Real-Time Prediction

The feature engineering module transforms raw API observations into an eight-dimensional feature vector comprising: temp_avg (average temperature), humidity_avg (average humidity), temp_range (temperature range, set to a default of 5°C when unavailable from the API), wind_speed, solar_radiation, cloud_cover, uv_index, and pressure_surface_level. These features were selected based on the EDA findings in Section IV-F, which identified temperature, humidity, and solar radiation as the strongest discriminating variables for heat intensity prediction.

D. Model Architecture

Two Random Forest models are trained on the historical Delhi weather dataset:

Logistic Regression: Used to categorise weather conditions into three actionable heat levels: Low, Medium, and High. This categorical output is useful for practical decision-making by urban planners and public health officials. Logistic Regression was selected because it performs efficiently on structured tabular datasets, provides fast training, and offers interpretability through feature coefficients that indicate the influence of variables such as temperature, humidity, and wind speed on heat risk levels.

Random Forest Classifier: Used to categorise the predicted conditions into three actionable heat levels: Low, Medium, and High. This categorical output is designed for practical decision-making by urban planners and public health officials. Random Forest was chosen over other classifiers because gradient-boosted tree ensembles consistently achieve competitive F1 scores on tabular meteorological classification tasks [9], and Random Forest provides robust performance with minimal hyperparameter tuning, natural feature importance rankings, and resistance to overfitting through bagging.

VI. PREPROCESSING AND MODELLING STRATEGY

A. Deduplication

The 2,431 duplicate (date, latitude, longitude) triplets are removed using a keep-first strategy before any feature engineering. Retaining duplicates would produce erroneously short lag windows and inflate accuracy estimates.

B. Train/Test Split

A strict time-based split is mandatory to prevent temporal leakage. The adopted split trains on records from January 2023 to September 2023 and tests on October 2023 to January 2024 (approximately 75%/25%). Random shuffling of rows is

explicitly prohibited, as it would allow future weather states to influence past predictions.

C. Imputation

Consistent with the MAR pattern and autocorrelated nature of weather series [14], the imputation strategy is: (1) forward-fill for gaps of ≤ 2 hours after sorting by timestamp; (2) monthly seasonal median imputation for longer gaps; and (3) addition of binary `is_missing_*` indicator columns. Global mean imputation is avoided as it distorts temporal autocorrelation and can cause leakage if computed over the full dataset.

D. Feature Engineering for Model Training

Calendar features (month-of-year, hour-of-day) encode seasonality and diurnal cycles. Lag features—rolling means of max_temperature, dew_point, and min_humidity over 3, 6, 12, and 24 h windows—capture thermal persistence and are constructed exclusively from past observations to prevent leakage [10]. A wet-bulb proxy feature derived from temperature and dew point [6] is used as an additional engineered predictor.

E. Normalisation

Random Forest models are invariant to monotonic feature transformations and do not require normalisation. This is an advantage over linear or neural network baselines that would require standardisation (zero mean, unit variance) computed on training data only.

F. Evaluation Metrics

Given the class imbalance (14.35% positive rate), primary evaluation metrics are PR-AUC and macro-F1; ROC-AUC is reported as a secondary metric. Accuracy alone is not used as a primary metric given the 85.65% majority-class baseline. For the regression model, Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) are reported. Time-series cross-validation with expanding windows is used to respect temporal ordering.

VII. RESULTS

A. Classification Performance

The Random Forest classifier, trained with class-weighted loss to address the 14.4% imbalance, demonstrates effective discrimination between heatwave and non-heatwave conditions. The model leverages the strong feature separability identified in the EDA—particularly the dominant predictive signal from max_temperature (Spearman $\rho \approx 0.51$) and the dry-heat interaction between temperature and humidity.

B. Regression Performance

The Random Forest regressor provides continuous heat intensity predictions that capture the non-linear threshold behaviour identified in Section IV-G, where heatwave probability transitions sharply near 40–45°C. The ensemble nature of Random Forest naturally models this non-linearity without requiring explicit feature transformations.

TABLE IV. SAMPLE PREDICTION RESULTS

Temp (°C)	Humidity (%)	Intensity	Category
32	55	25.3	Medium
36	40	30.8	High
28	70	18.2	Low
42	30	38.5	High
35	50	28.1	Medium

C. Dashboard Capabilities

The Streamlit-based interactive dashboard provides: real-time trend graphs showing temperature and heat intensity over time; heat intensity visualisation with colour-coded severity indicators; historical tracking allowing users to examine past patterns; and a map-based location selector for querying any point within the Delhi metropolitan region. The dashboard enables non-technical stakeholders—urban planners, public health officials, and emergency responders—to access heat intensity predictions without requiring ML expertise.

VIII. LIMITATIONS

Several limitations constrain the scope of this work:

First, the most critical limitation is label validity: heatwave streaks of median 1 hour and maximum 3 hours are inconsistent with IMD and WMO definitions requiring multi-day persistence. Any model trained on this label predicts instantaneous heat-stress exceedance, not a canonical heatwave. Relabelling using a persistence-based criterion (e.g., $\text{max_temperature} \geq 40^\circ\text{C}$ for ≥ 48 consecutive hours, departing from normal by $\geq 4.5^\circ\text{C}$) is necessary before operational use.

Second, the dataset covers only 13 months (January 2023–January 2024), limiting representation of inter-annual variability. Third, the station identification by coordinate pairs only—with no metadata on sensor type, height, or land-cover context—prevents rigorous quality control and urban heat island correction. Fourth, spatial information (atmospheric circulation patterns, teleconnections) is absent from the point-level data. Fifth, the real-time prediction system uses approximate solar radiation and UV index values from the OpenWeatherMap API, which may differ from ground-truth sensor measurements. Sixth, the current system is limited to single-point predictions and does not generate a full spatial heatmap across Delhi.

IX. CONCLUSION AND FUTURE WORK

This paper presented a comprehensive study of urban heat intensity prediction for Delhi, integrating systematic EDA, literature review, and machine learning implementation. From the analysis of 9,505 hourly records, five critical EDA findings emerge: (1) moderate class imbalance at 14.35% requiring PR-AUC and macro-F1 evaluation; (2) strong feature separability with max_temperature ($\rho \approx 0.51$, $\text{SMD} = +1.18$) and min_temperature ($\text{SMD} = +1.32$) as dominant predictors and min_humidity ($\text{SMD} = -0.83$) confirming the dry-heat paradigm; (3) approximately 10–11% uniform MAR missingness handled through time-aware imputation; (4) 2,431

duplicate timestamps removed before modelling; and (5) heatwave streaks inconsistent with standard multi-day definitions, requiring label reconstruction.

Building on these findings, the implemented Random Forest-based prediction system successfully predicts both continuous heat intensity and categorical heat levels using real-time weather data, with an interactive Streamlit dashboard enabling practical use by non-technical stakeholders. The system demonstrates that integrating ML models with live weather APIs and interactive interfaces can bridge the gap between research and operational deployment for urban heat monitoring.

Future work should pursue five directions: (1) reconstructing the target label using a persistence-based IMD-compliant criterion before further model development; (2) extending coverage to a full Delhi spatial heatmap using multiple sensor locations or satellite-derived data; (3) developing a mobile application for broader accessibility; (4) implementing an automated alert system that notifies public health agencies when predicted heat intensity exceeds critical thresholds; and (5) incorporating gridded reanalysis data for spatial feature extraction and validating against independent IMD ground-truth heatwave declarations.

TABLE V. AUTHOR CONTRIBUTIONS

Author	Roll No.	Contributions
Abhishek Kumar	24293916112	Dataset preprocessing; ML Model Selection; EDA Graphs
Pavan Agrawal	24293916041	ML Model Implementation and Dashboard
Manav Bhatia	24293916122	Dataset Sourcing; SMD computation; Feature separability analysis
Harsh Nath Tripathi	24293916101	Feature engineering roadmap; Missing Data Analysis and Imputation

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